HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Windows Security Event IDs for Hunters

The events that matter on Windows 11 and Server 2022/2025, the fields that carry the signal, the codes you decode mid-hunt, and the audit policy that has to be on first.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-11-03

Events marked **(audit)** are silent until the matching Advanced Audit Policy subcategory is enabled; the Enable it section at the end lists them. Field names are the XML `EventData` names (what Sentinel's `SecurityEvent` and Splunk's XmlWinEventLog use). See Data & telemetry for the collection side.

Authentication and logon

ID	Meaning	Hunting use (ATT&CK)
4624	Successful logon	The workhorse. Read `LogonType`, `IpAddress`, `WorkstationName`, `AuthenticationPackageName` (NTLM/Kerberos/Negotiate), `LogonProcessName` (`NtLmSsp`, `Kerberos`, `User32`, `seclogo` = runas/pth), `ElevatedToken` (`%%1842` yes, `%%1843` no), `TargetLogonId` to tie later activity to the session. Lateral movement, account misuse (T1021, T1078)
4625	Failed logon	Brute force and spray via `Status` / `SubStatus` (table below), `LogonType`, `IpAddress`. 4625 is logged on the machine that was targeted; sprays against DCs over NTLM show as 4776 instead (T1110)
4634 / 4647	Logoff / user-initiated logoff	Pair with 4624 on `TargetLogonId` for session duration; very short type-3 sessions in bulk = tooling
4648	Logon with explicit credentials	`runas`, PsExec `-u`, scheduled tasks under another identity, overpass-the-hash. `TargetUserName` is the account used, `SubjectUserName` who used it; a non-admin presenting admin creds is the signal (T1078, T1550)
4672	Special privileges assigned to new logon	Fires with every admin-equivalent 4624 (`SeDebugPrivilege`, `SeTcbPrivilege`, `SeBackupPrivilege` ...). Baseline admin logons per host; first-time 4672 for an account = escalation (T1078.002)
4776	NTLM credential validation on the DC (or local SAM)	Where NTLM sprays and pass-the-hash land: `Workstation` is the source, `Status` carries the failure code; NTLM where Kerberos is expected (T1550.002, T1110)
4768	Kerberos TGT requested (AS-REQ)	`PreAuthType` 0 with `TicketEncryptionType` 0x17 = AS-REP roasting; `CertIssuerName` / `CertThumbprint` populated = PKINIT (AD CS or shadow-credential logon); unusual `IpAddress` for privileged accounts (T1558.004, T1649)
4769	Kerberos service ticket requested (TGS-REQ)	Kerberoasting: one `TargetUserName` requesting many `ServiceName` values with 0x17; silver-ticket hunts compare 4769 on the DC with 4624 on the service host (T1558.003, T1558.002)
4770	Service ticket renewed	Normal; abnormal renewal patterns hint at long-lived forged tickets
4771	Kerberos pre-authentication failed	Kerberos spray and lockout investigations: `Status` 0x18 bad password, 0x12 disabled/locked, `IpAddress` source (T1110)
4778 / 4779	RDP session reconnected / disconnected	`ClientName`, `ClientAddress` for RDP session tracing; combines with 4624 type 10 (T1021.001)
4800 / 4801	Workstation locked / unlocked	User presence; activity during a locked session is not the user
4825	RDP access denied (not in Remote Desktop Users)	Lateral-movement probing with low-priv accounts
4964	Special groups assigned to a new logon	Fires only if you configure Special Groups; cheap tripwire for tier-0 logons

Logon types (4624 / 4625 / 4634)

Type	Name	What produces it
2	Interactive	Console or VNC-style keyboard logon

Type	Name	What produces it
3	Network	SMB, WinRM, WMI, RPC, LDAP, IIS Windows auth; most lateral movement
4	Batch	Scheduled tasks
5	Service	Service start under an account
7	Unlock	Workstation unlock
8	NetworkCleartext	Basic auth over IIS or PowerShell `-Authentication Basic`; credential exposure
9	NewCredentials	`runas /netonly`, Mimikatz `sekurlsa::pth`, Rubeus `createnetonly`; `LogonProcessName` is `seclogo`
10	RemoteInteractive	RDP, Remote Assistance
11	CachedInteractive	Domain logon with cached credentials, DC unreachable
12 / 13	CachedRemoteInteractive / CachedUnlock	Cached variants of 10 and 7

Failure codes (4625 `Status` / `SubStatus`, 4776 `Status`)

Code	Meaning	Read as
`0xC000006D`	Bad username or authentication information (wrapper)	Look at `SubStatus`
`0xC000006A`	Correct username, wrong password	Spray or brute force
`0xC0000064`	Username does not exist	Enumeration, spray with a guessed list
`0xC0000234`	Account locked out	Spray aftermath
`0xC0000072`	Account disabled	Attempts against stale accounts
`0xC000006F` / `0xC0000070`	Outside logon hours / workstation restriction	Policy, or an attacker using a restricted account
`0xC0000193` / `0xC0000071`	Account expired / password expired	Noise, unless clustered
`0xC000015B`	Logon type not granted	Service account used interactively, or blocked RDP
`0xC0000133`	Clock skew	Kerberos from a host with a bad clock, or a forged ticket
`0xC000035B`	NTLM authentication failed (SPN/MIC validation)	Relay attempts against hardened targets

Kerberos fields (4768 / 4769 / 4771)

Field	Values	Note
`TicketEncryptionType`	`0x12` AES256, `0x11` AES128, `0x17` RC4-HMAC, `0x18` RC4-HMAC-EXP, `0x3` DES-CBC-MD5, `0x1` DES-CBC-CRC, `0xFFFFFFFF` on failure	Modern estates are AES; 0x17 for a user account is the roasting tell, DES means an ancient client or a forged ticket
`PreAuthType` (4768)	`0` none, `2` PA-ENC-TIMESTAMP (password), `11` PA-ETYPE-INFO, `15` PA-PK-AS-REP_OLD (older smart-card logon), `16` PA-PK-AS-REQ (PKINIT, certificate), `17` PA-PK-AS-REP, `138` PA-ENCRYPTED-CHALLENGE (FAST armouring)	`0` + success = AS-REP roastable account; `15` / `16` + an unusual `CertIssuerName` = certificate abuse (AD CS, shadow credentials)
`TicketOptions`	hex bitmask: `0x40000000` forwardable, `0x20000000` forwarded, `0x00800000` renewable, `0x00010000` canonicalize, `0x00000010` renewable-ok; Windows clients mostly send `0x40810010` or `0x40810000`	Uncommon combinations (for example `0x40800000`, `0x50800000`) are tool fingerprints; treat as a lead only
`Status` / `ResultCode` (4768, 4771, 4769)	`0x6` client not found (unknown user), `0x7` server not found (bad SPN), `0x12` client revoked (disabled or locked), `0x17` password expired, `0x18` pre-auth failed (bad password), `0x1F` integrity check failed, `0x20` ticket expired, `0x25` clock skew too great	`0x6` spikes = user enumeration (Kerbrute), `0x18` spikes = Kerberos spray, `0x25` from one host = forged ticket or broken clock

Process, scheduled tasks and services

ID	Meaning	Hunting use (ATT&CK)
4688	Process created **(audit; enable command line)**	`NewProcessName`, `CommandLine`, `ParentProcessName`, `SubjectUserName`, `TokenElevationType` (`%%1936` full/UAC-off, `%%1937` elevated, `%%1938` limited), `MandatoryLabel`. Parent/child chains and LOLBins when Sysmon is absent (T1059, T1218). Without `CommandLine` it is nearly useless
4689	Process exited	`ExitStatus`; pair on `ProcessId` for lifetime of short-lived tools
4696	Primary token assigned to process	Token manipulation, rare (T1134)
4697	Service installed in the system **(audit: Security System Extension)**	`ServiceName`, `ServiceFileName`, `ServiceType`, `ServiceStartType`; PsExec-style services, binary paths in user-writable dirs (T1543.003, T1569.002)
7045 (System)	New service installed	Same signal with no audit policy needed; `ImagePath`, `AccountName`; `cmd.exe /c` or `powershell` in the path = remote execution
7034 / 7036 / 7040 (System)	Service crashed / entered state / start type changed	Defender or EDR service stopped or set to disabled (T1562.001); LSASS crash after a dump attempt
4698 / 4699	Scheduled task created / deleted **(audit: Other Object Access)**	`TaskName`, `TaskContent` (full XML: command, arguments, trigger, principal); creation then deletion within minutes = one-shot remote execution (T1053.005)
4700 / 4701 / 4702	Task enabled / disabled / updated	Updated tasks that now point at a new binary; disabled security tasks
106 / 140 / 141 / 200 / 201 (TaskScheduler/Operational)	Task registered / updated / deleted / action started / action completed	Covers hosts without the Security audit; 200 names the exact executable that ran

Accounts and groups

ID	Meaning	Hunting use (ATT&CK)
4720	User account created	`SamAccountName`, `UserAccountControl`, `SubjectUserName`; creation by a non-helpdesk account or on a workstation (T1136.001/.002)
4722 / 4725	Account enabled / disabled	Re-enabled stale accounts
4723 / 4724	Password change attempt (by user) / reset attempt (by admin)	Resets of privileged or service accounts outside change windows
4726	Account deleted	Cleanup after use
4738	User account changed	`UserAccountControl` deltas: pre-auth disabled (`%%2096`), password not required (`%%2082`), DES only, trusted for delegation (`%%2093`); `AllowedToDelegateTo`, `SidHistory`, `PrimaryGroupId` (T1098, T1558.004)
4740 / 4767	Account locked out / unlocked	`TargetUserName`, `TargetDomainName` = source host of the lockout
4728 / 4729	Member added / removed: security-enabled global group	Domain Admins is global: watch here (T1098)
4732 / 4733	Member added / removed: security-enabled local group	Local Administrators, Remote Desktop Users, Backup Operators
4756 / 4757	Member added / removed: security-enabled universal group	Enterprise Admins, Schema Admins
4727 / 4730 / 4731 / 4734 / 4754 / 4758	Group created / deleted	New groups that mimic admin names
4741 / 4742 / 4743	Computer account created / changed / deleted	Machine-account quota abuse (any user creating computers), `msDS-AllowedToActOnBehalfOfOtherIdentity` set (RBCD), `UserAccountControl` or `ServicePrincipalNames` changes on DC accounts (T1136.002, T1550)

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ID	Meaning	Hunting use (ATT&CK)
4781	Account name changed	sAMAccountName spoofing: computer renamed to a DC name without `$` (CVE-2021-42278 pattern)
4794	DSRM administrator password set attempt	Persistence through the DSRM account; should be zero outside DC builds (T1098)
4765 / 4766	SID History added / add attempt failed	SIDHistory injection (T1134.005)

Directory service and Active Directory (DCs)

ID	Meaning	Hunting use (ATT&CK)
4662	Operation performed on an AD object **(audit: Directory Service Access + SACL)**	DCSync: `Properties` containing `1131f6aa-9c07-11d1-f79f-00c04fc2dcd2` (DS-Replication-Get-Changes), `1131f6ad-...` (Get-Changes-All) or `89e95b76-444d-4c62-991a-0facbeda640c` (Get-Changes-In-Filtered-Set) from a `SubjectUserName` that is not a DC or Entra Connect account (T1003.006). Also reads of `msDS-ManagedPassword` (gMSA) and LAPS attributes
5136	Directory object modified **(audit: Directory Service Changes)**	`AttributeLDAPDisplayName` is the hunt key: `msDS-KeyCredentialLink` (shadow credentials, T1556), `msDS-AllowedToActOnBehalfOfOtherIdentity` (RBCD), `nTSecurityDescriptor` on AdminSDHolder or privileged objects (T1098), `gPCFileSysPath` / `versionNumber` on `groupPolicyContainer` (T1484.001), `servicePrincipalName` on users (targeted Kerberoast), `scriptPath`, `msDS-SupportedEncryptionTypes`
5137 / 5141	Directory object created / deleted	New objects in Configuration NC (`nTDSDSA`, `server` = DCShadow, T1207), new OUs or GPOs, deleted objects after abuse
5138 / 5139	Object undeleted / moved	Moves into less-audited OUs
4670	Permissions on an object changed	`ObjectName`, `OldSd` / `NewSd` SDDL; new ACEs on tier-0 objects, GenericAll/WriteDACL grants (T1222)
4928 / 4929	Replica source naming context established / removed	Rogue DC registration (DCShadow)
4932 / 4933	Replication started / ended	Replication from a source that is not a DC
4886 / 4887 / 4888 (AD CS, on the CA)	Certificate request received / approved and issued / denied	`Requester`, `Attributes` (look for `SAN:` or `upn=` for another user: ESC1), `CertificateTemplate`; certs for privileged UPNs requested from non-tier-0 hosts (T1649)
4889 / 4890 / 4891 / 4892	Request set to pending / CA security changed / CA configuration changed / CA property changed	`EDITF_ATTRIBUTESUBJECTALTNAME2` flips (ESC6), new CA officers
4896 / 4897	Rows deleted from the CA database / role separation enabled	Cleanup of issued-cert evidence
4898 / 4899 / 4900	Template loaded / template updated / template security updated	Template edits that add Client Authentication EKU, enable enrollee-supplies-subject, or widen enrol rights (ESC4)
1644 (Directory Service)	Expensive or inefficient LDAP search	Bulk LDAP recon (BloodHound, ADFind): `Client`, `Filter`, `Visited entries`. Enable with `Field Engineering = 5` and `Expensive Search Results Threshold` / `Inefficient Search Results Threshold` under `HKLM\SYSTEM\CurrentControlSet\Services\NTDS\Parameters` (T1087.002, T1482)
2889 (Directory Service)	Unsigned or simple LDAP bind	Relay-prone clients; `Client IP`, `Identity`
5827 / 5828 / 5829 (System, Netlogon)	Netlogon secure channel denied / vulnerable connection allowed	Zerologon-style abuse and machines still using RC4 secure channels (T1210)

Network shares and object access

ID	Meaning	Hunting use (ATT&CK)
5140	Network share accessed **(audit: File Share)**	`ShareName` (`\\*\ADMIN$` , `\\*\C$` , `\\*\IPC$` , `\\*\SYSVOL`), `IpAddress` , `SubjectUserName` , `AccessMask` ; one account touching many hosts' admin shares (T1021.002, T1135)
5145	Share object access checked **(audit: Detailed File Share)**	`RelativeTargetName` gives the file: `PSEXESVC.exe` , `*.exe` written to `ADMIN$` , `svcctl` / `srvsvc` / `samr` / `lsarpc` / `winreg` pipes over `IPC$` , `ntds.dit` on backup shares, reads of `.kdbx` / `id_rsa` / `web.config` (T1021.002, T1039). High volume on file servers, filter by `AccessMask` 0x2 (write) or by sensitive names
5142 / 5143 / 5144	Share created / modified / deleted	Staging shares, exposed `C$` -like shares on workstations
5168	SPN check for SMB/SMB2 failed	NTLM relay to SMB with signing (lead)
4656 / 4658 / 4663	Handle requested / handle closed / object accessed **(audit: File System, Registry, Kernel Object + SACL on the object)**	SACL-driven: LSASS handle requests (`ObjectName` ends `lsass.exe` , `AccessMask` 0x1010/0x1FFFFF), reads of honey files, SAM/SECURITY hive access (T1003). Nothing is logged without a SACL
4657	Registry value modified **(audit: Registry + SACL)**	Run keys, `Image File Execution Options` , Winlogon, services `ImagePath` , Defender exclusions (T1547.001, T1546.012, T1562.001)
4670	Permissions on an object changed	File or registry ACL loosened before abuse
4660	Object deleted	Paired with 4663 for evidence removal
5379 / 5381 / 5382	Credential Manager credentials read / Vault credentials read (two sources)	`cmdkey /list` , Mimikatz `vault::cred` , browser and RDP credential theft via DPAPI (T1555.004)
4673 / 4674	Privileged service called / operation on a privileged object	`SeDebugPrivilege` use by non-debuggers, `SeTcbPrivilege` ; noisy, filter by privilege and process (T1134)

Policy, log and system integrity

ID	Meaning	Hunting use (ATT&CK)
1102	Security log cleared	Always investigate; `SubjectUserName` is who did it (T1070.001)
104 (System)	Other event log cleared	System/Application/PowerShell logs wiped
1100	Event logging service shut down	Outside clean shutdowns = tamper
1104 / 1105	Security log full / automatic backup	Log flooding to roll evidence
4719	System audit policy changed	`SubcategoryGuid` , `AuditPolicyChanges` (`%%8448` success removed, `%%8450` failure removed); auditing switched off before action (T1562.002)
4817	Auditing settings on an object changed	SACL removed from LSASS or a honey object
4904 / 4905	Security event source registered / unregistered	Rare; source tampering
4616	System time changed	`PreviousTime` / `NewTime` , `ProcessName` ; timestomping of logs, Kerberos manipulation (T1070.006)
4826	Boot configuration data loaded	`TestSigning` , `DisableIntegrityChecks` , `HypervisorDebug` ; BYOVD and Safe-Mode tricks (T1553, T1562)
4697 / 4702 / 4719 / 1102 within minutes	Classic evasion sequence	Stack by host
6416	New external device recognised	`DeviceDescription` , `ClassName` ; USB storage and HID devices on servers (T1091, T1200)
5038 / 6281	Code integrity: image hash invalid / page hashes invalid	Tampered or unsigned system files, bootkits (T1014)
4608 / 4609	Windows starting up / shutting down	Reboot after persistence install

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Other logs worth ingesting

Log (channel)	IDs	Why
`Microsoft-Windows-PowerShell/Operational`	**4104** script block (content, `ScriptBlockText`, `Path`), **4103** module logging (parameters), **4105/4106** script start/stop; classic **400/403** engine state, **800** pipeline execution in `Windows PowerShell`	Encoded commands, AMSI bypass strings, download cradles; 4104 level Warning fires automatically for suspicious blocks even without the policy (T1059.001). Enable Script Block Logging and Module Logging; transcription to a write-only share for full text
`Microsoft-Windows-WinRM/Operational`	**91** session created (server), **142** request failed, **168** session created by user, **169** user authenticated, **6** WSMan session init (client)	Remote PowerShell: pair 91/168 on the target with 4104 on the target and 4624 type 3 `wsmprovhost.exe` children (T1021.006)
`Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational`	**1149** network-layer authentication succeeded	Earliest RDP evidence with `Source Network Address`, fires before 4624 (T1021.001)
`Microsoft-Windows-TerminalServices-LocalSessionManager/Operational`	**21** session logon, **22** shell start, **23** logoff, **24** disconnected, **25** reconnected, **39/40** session arbitration	RDP session timeline per `Session ID` and source address
`Microsoft-Windows-TaskScheduler/Operational`	106, 140, 141, 200, 201	See above; off by default on Server, enable the channel
`Microsoft-Windows-WMI-Activity/Operational`	**5857** provider loaded, **5858** query error, **5859** permanent subscription started, **5860** temporary subscription, **5861** permanent subscription created (shows filter and consumer)	WMI persistence and remote WMI execution; 5861 is the one to alert on (T1546.003, T1047)
`Microsoft-Windows-Windows Defender/Operational`	**1006/1007/1015** malware detected / action taken / suspicious behaviour, **1116/1117/1118/1119** detection / action succeeded / failed / critically failed, **5001** real-time protection disabled, **5004** RTP configuration changed, **5007** configuration changed (exclusions), **5010/5012** scanning for malware/scripts disabled, **5013** tamper protection blocked a change, **1121/1122** ASR block/audit, **2050/2051** sample submission	Defender tampering and the attacks it caught but you ignored (T1562.001)
`Microsoft-Windows-Windows Firewall With Advanced Security/Firewall`	**2004** rule added, **2005** rule modified, **2006** rule deleted, **2033** all rules deleted, **4946-4950** (Security)	New allow rules for RDP, SMB or a C2 port (T1562.004)
`Microsoft-Windows-AppLocker/EXE and DLL`, `/MSI and Script`, `/Packaged app-*`	**8002** allowed, **8003** audit-would-block, **8004** blocked, **8005/8006/8007** script allowed/audit/blocked	Audit-mode output is a free execution inventory (T1218, T1059)
`Microsoft-Windows-CodeIntegrity/Operational` (WDAC, LSA protection)	**3076** audit: would be blocked, **3077** blocked, **3089** signature info, **3033** driver or process blocked by policy, **3065/3066** LSA protection: plug-in failed signing requirements (audit/enforce)	Unsigned drivers and binaries, BYOVD, attempts to load code into protected LSASS (T1068, T1553.006, T1003.001)
`Microsoft-Windows-NTLM/Operational`	**8001** outgoing NTLM to a remote server, **8002** incoming NTLM, **8003** NTLM to the domain, **8004** NTLM audited/blocked by policy	Requires `Restrict NTLM: Audit...` policies; relay, PtH and legacy-protocol inventory (T1557.001, T1550.002)

Log (channel)	IDs	Why
`Microsoft-Windows-Bits-Client/Operational`	**3** job created, **59** transfer started (URL), **60** transfer completed, **61** transfer failed, **4** job completed	`bitsadmin` / `Start-BitsTransfer` downloads with the full URL (T1197, T1105)
`Microsoft-Windows-DNS-Client/Operational`	**3008** query completed (`QueryName` , `QueryResults`), **3020** response	Process-less DNS when Sysmon 22 is absent; analytic channel must be enabled and is noisy (T1071.004)
`Microsoft-Windows-Kernel-PnP/Configuration`	**400/410** device configured / started	USB device history with VID/PID, complements 6416 (T1091)
`Microsoft-Windows-Security-Mitigations/KernelMode` and `/UserMode`	**10-23** exploit-protection blocks (ACG, CFG, child-process, image load)	Exploit attempts and injection blocked by Exploit Protection (T1055, T1203)
`Microsoft-Windows-Sysmon/Operational`	1-29	See the Sysmon sheet

Quick hunts by event

Hunt	Events and query idea	Playbook
Password spray	4625 with `SubStatus` 0xC000006A or 0xC0000064, one `IpAddress` , many `TargetUserName` , few tries each; 4771 0x18 and 4776 on DCs for the Kerberos/NTLM variants	Cloud & identity (spray section), Lateral movement
Pass-the-hash	4624 `LogonType` 3, `AuthenticationPackageName` NTLM, `KeyLength` 0, `LogonProcessName` `NtLmSsp` from a workstation to many servers; 4624 type 9 `seclogo` on the source	Lateral movement
Kerberoasting	4769 `TicketEncryptionType` 0x17, `ServiceName` not ending `$` , dcount(`ServiceName`) per `TargetUserName` per 10 min	Kerberoasting
AS-REP roasting	4768 `PreAuthType` 0, 0x17, `TargetUserName` with `DONT_REQ_PREAUTH`	Kerberoasting
DCSync	4662 `Properties` has `1131f6aa` or `1131f6ad` , `SubjectUserName` not a DC `$` account or Entra Connect	DCSync
Shadow credentials / RBCD	5136 `AttributeLDAPDisplayName` in (`msDS-KeyCredentialLink` , `msDS-AllowedToActOnBehalfOfOtherIdentity`) by a non-sync account, then 4768 `PreAuthType` 15/16	AD CS abuse, Lateral movement
ESC1 certificate abuse	4886/4887 `Attributes` contains `SAN:` or `upn=` for a user other than `Requester` , template with enrollee-supplies-subject	AD CS abuse
PsExec-style remote execution	5145 `ShareName` `ADMIN$` , `RelativeTargetName` `*.exe` , `AccessMask` 0x2, then 7045/4697 on the target, service child of `services.exe`	Lateral movement
Remote scheduled task	4698 with `TaskContent` command in a user-writable path, `SubjectLogonId` from a type-3 session; 4699 within minutes	Persistence
WMI persistence	5861 with `CommandLineEventConsumer` or `ActiveScriptEventConsumer` ; 5859/5860 bursts	Persistence
RDP from the inside	1149 then 21/22 with `Source Network Address` internal, stack by source; 4778/4779 reconnects at odd hours	RDP & WinRM anomalies
WinRM lateral movement	91/168 on the target, 4624 type 3 with `wsmprovhost.exe` children in 4688, 4104 on the target	RDP & WinRM anomalies
Defender tampering	5001, 5007 (`Exclusions\Paths` or `Processes` added), 5010/5012, 5013 blocked attempts; 7036 Defender service stopped	Defense evasion
Log clearing and audit tampering	1102, 104, 1100, 4719 with `%%8448` / `%%8450` , 4817 on LSASS or honey objects	Defense evasion

Hunt	Events and query idea	Playbook
Privilege group changes	4728/4732/4756 where `TargetUserName` is an admin group; 4672 first-seen per account	AD discovery & BloodHound (what precedes it)
Ransomware precursors	7036 for backup/AV services stopping, 4697 new services from `C:\Users` or `C:\ProgramData`, 5140 bursts across many hosts by one account	Ransomware precursors

Enable it

Nothing above fires on a default install except the basics. Set the **Advanced Audit Policy** (Computer Configuration > Policies > Windows Settings > Security Settings > Advanced Audit Policy Configuration) and force it with *Audit: Force audit policy subcategory settings to override audit policy category settings*.

Section above	Subcategories (Success unless noted)
Authentication and logon	Logon/Logoff: Logon (S+F), Logoff, Special Logon, Other Logon/Logoff Events; Account Logon: Credential Validation (S+F), Kerberos Authentication Service (S+F), Kerberos Service Ticket Operations (S+F) on DCs
Process, tasks, services	Detailed Tracking: Process Creation, Process Termination; Object Access: Other Object Access Events (tasks); System: Security System Extension (4697)
Accounts and groups	Account Management: User Account Management, Computer Account Management, Security Group Management, Other Account Management Events
Directory service	DS Access: Directory Service Access (4662; needs SACLs, default SACL covers replication), Directory Service Changes (5136-5141)
Shares and object access	Object Access: File Share, Detailed File Share (5145, heavy), File System, Registry, Kernel Object (LSASS), Handle Manipulation off (too noisy)
Policy and integrity	Policy Change: Audit Policy Change, Authentication Policy Change, Authorization Policy Change; System: Security State Change, System Integrity, Other System Events

- **Command line in 4688**: Computer Configuration > Administrative Templates > System > Audit Process Creation > *Include command line in process creation events* = Enabled (`HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Audit\ProcessCreationIncludeCmdLine_Enabled = 1`). 4688 without it tells you `cmd.exe` ran and nothing else.
- **PowerShell**: Administrative Templates > Windows Components > Windows PowerShell: *Turn on Module Logging* (`*`), *Turn on PowerShell Script Block Logging*, *Turn on PowerShell Transcription*; PowerShell 7 reads the same policies from its own key.
- **Sizes**: Security log 1 GB or more on DCs and servers, 256 MB on workstations; forward with WEF or an agent rather than relying on local retention.
- **Verify**: `auditpol /get /category:*` on a sample of hosts, then confirm the events actually arrive in the SIEM (a query on a missing log is not a negative result).

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